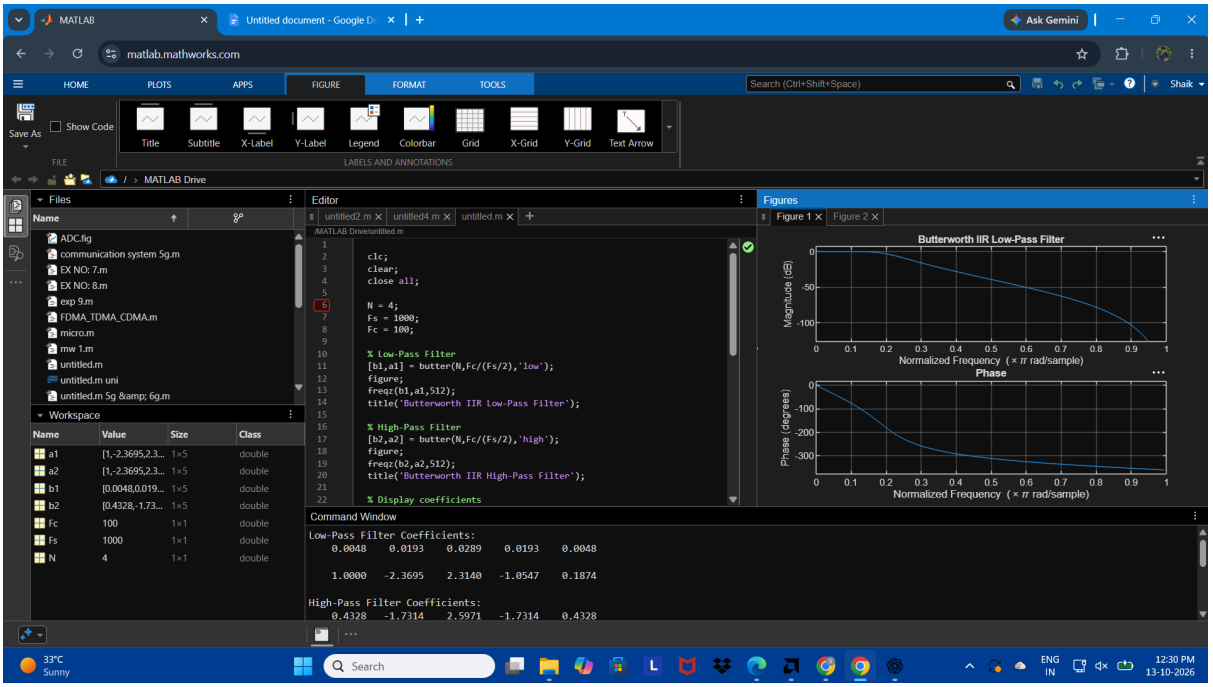
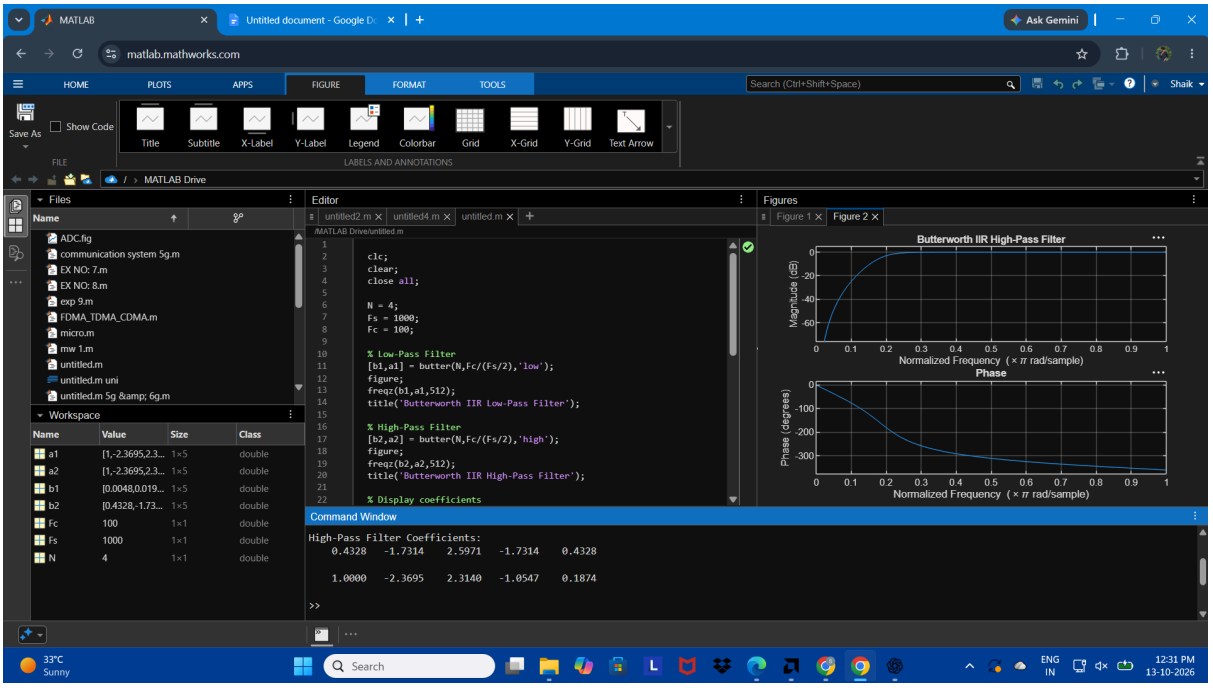
EXP NO:- 1

PROGRAM ➖

```
clc;
clear;
close all;
N = 4;
Fs = 1000;
Fc = 100;
% Low-Pass Filter
[b1,a1] = butter(N,Fc/(Fs/2),'low');
figure;
freqz(b1,a1,512);
title('Butterworth IIR Low-Pass Filter');
% High-Pass Filter
[b2,a2] = butter(N,Fc/(Fs/2),'high');
figure;
freqz(b2,a2,512);
title('Butterworth IIR High-Pass Filter');
% Display coefficients
disp('Low-Pass Filter Coefficients:');
disp(b1);
disp(a1);
disp('High-Pass Filter Coefficients:');
disp(b2);
disp(a2);
```

OUTPUT:-


Butterworth IIR Low-Pass Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
Low-Pass Filter Coefficients:
0.0048 0.0193 0.0289 0.0193 0.0048
1.0000 -2.3695 2.3140 -1.0547 0.1874
High-Pass Filter Coefficients:
0.4328 -1.7314 2.5971 -1.7314 0.4328


Butterworth IIR High-Pass Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
High-Pass Filter Coefficients:
0.4328 -1.7314 2.5971 -1.7314 0.4328
1.0000 -2.3695 2.3140 -1.0547 0.1874